



Effects of process parameters on the surface characteristics of laser powder bed fusion printed parts: machine learning predictions with random forest and support vector regression

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Abstract

Laser powder bed fusion (L-PBF) fuses metallic powder using a high-energy laser beam, forming parts layer by layer. This technique offers flexibility and design freedom in metal additive manufacturing (MAM). However, achieving the desired surface quality remains challenging and impacts functionality and reliability. L-PBF process parameters significantly influence surface roughness. Identifying the most critical factors among numerous parameters is essential for improving quality. This study examines the effects of key process parameters on the surface roughness of AlSi10Mg, a widely used aluminum alloy in high-tech industries, fabricated by L-PBF. Part orientation, laser power, scanning speed, and layer thickness were identified as crucial parameters via cause-and-effect analysis. To systematically examine their effects, the Taguchi method was employed within the framework of the design of experiment (DoE). Experimental results and statistical analysis revealed that laser power, scanning speed, and layer thickness significantly influence surface roughness parameters: arithmetic mean (Ra) and root mean square (Rq). Main effect plots and energy density analyses confirmed their impact on surface quality. Microscopic investigations identified surface flaws such as spattering, balling, and porosity contributing to poor quality. Given the complex interplay between parameters and surface quality, accurately predicting their effects is challenging. To address this, machine learning models, specifically random forest regression (RFR) and support vector regression (SVR), were used to predict the effects on surface roughness. The RFR model's R^2 values for predicting Ra and Rq are 97% and 85%, while the SVR model's predictions are 85% and 66%, respectively. Evaluation metrics demonstrated that the RFR model outperformed SVR in predicting surface roughness.

Keywords Additive manufacturing · Laser powder bed fusion · Machine learning · Surface roughness · Support vector regression · Random forest regression

1 Introduction

Additive manufacturing (AM) technology has revolutionized the modern manufacturing landscape integrating graphic design, digital control, and welding, thereby reducing both production time and costs [1]. With computer-aided design (CAD), products are designed, modeled, and generated to sliced format [2]. Among metal additive manufacturing (MAM) technologies, particularly powder bed fusion (PBF) uses either a laser or electron beam as an energy source to melt and fuse metal (or ceramic) powders [3]. Laser powder bed fusion (L-PBF), as one of these techniques, stands out due to its promising capabilities, including manufacturing flexibility, complex shape design capability, and savings in production time and costs [4–6]. It also excels in producing thin-walled parts with excellent strength-to-weight ratios,

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such as thin-walled tubes or channels [7–10]. Moreover, L-PBF offers superior advantages over traditional methods for manufacturing complex structures like the Gyroid variant of triply periodic minimal surface (TPMS) [11]. It enables precise fabrication, crucial for aerospace, automotive, and medical applications, by directly translating digital designs into physical objects layer by layer. This approach enhances design flexibility, reduces material waste, and shortens manufacturing lead times, thus driving the optimization of structures across industries.

L-PBF is regarded as a potential novel AM technology [7] and finds applications in various industries such as jewelry, automotive, oil refinery, marine, construction, aerospace, and aviation [12]. However, despite its numerous advantages, L-PBF faces challenges in achieving satisfactory surface quality, particularly in industrial applications, where surface roughness significantly impacts component performance. According to Boschetto et al. [13], surface roughness in L-PBF can vary within the range of 5 to 40 μm , which is lower than other metal-based AM techniques such as electron beam melting (25–130 μm) [14] and wire arc AM, where the surface roughness measurements often exceed several hundred micrometers [15]. While L-PBF has the capacity to achieve the best surface quality among metal-based AM, it frequently falls short of meeting industrial standards.

Surface roughness is a common challenge in AM, particularly in L-PBF processes, and is widely recognized as a cause of premature failures [15, 16]. It is a critical quality determinant in L-PBF-produced parts, influencing various aspects such as tribological properties, fatigue performance, and overall mechanical integrity [5, 14, 17]. Studies investigating the impact of surface roughness on the fatigue life of Ti6Al4V parts produced by L-PBF highlight early fractures, poor mechanical properties, and compromised fatigue performance [18–21]. Currently, the surface roughness of AlSi10Mg alloy parts produced by L-PBF is a major concern in the aerospace and automotive industries, as it directly affects fatigue life and corrosion resistance [21]. Poor surface finish can lead to increased friction, wear, and susceptibility to crack initiation, ultimately affecting the lifespan and reliability of components. Studies on the relationship between arithmetic surface roughness (R_a) and wear resistance reveal a decrease in volumetric wear rate as the R_a value decreases [22]. The correlation between surface peaks (R_p), valleys (R_v), and corrosion resistance further underlines the interaction between surface conditions and material performance [22–24]. The impact of surface roughness on general corrosion, metastable pitting formation, and pitting potential adds layers of complexity to achieving the best surface quality, particularly given the interdependent nature of L-PBF process parameters [25–27]. Poor surface roughness has several detrimental effects, including reduced thermal

conductivity [28], increased risk of crack initiation [29], lower electrical conductivity [30], and decreased effectiveness of lubrication [31]. The impact of surface roughness on the performance of final products is illustrated in Fig. 1. To mitigate surface roughness issues, post-processing methods such as machining, shot peening, and chemical processing are employed. However, post-processing the inner surfaces of complex geometries remains challenging, and the associated costs often negate the advantages gained through AM [13, 27, 28]. The ideal of producing operational L-PBF parts without post-processing remains challenging, especially for applications such as aeronautics [32].

The prominent techniques were employed to study the influential process parameters with respect to the desired quality characteristics like categorizing the source of parameters and following the cause-and-effect analysis. In L-PBF, the sources of process parameters can be categorized into laser parameters, powder characteristics, feedstock material properties, and printing chamber environment [33]. Finding the responsible process parameters is crucial. Cause-and-effect analysis emerges as a valuable tool in this quest [34].

L-PBF emerges as a promising approach due to its ability to produce complex geometries with minimal waste [35], but challenges such as parameter dependency and material characteristics exist. AlSi10Mg, a widely used cast aluminum alloy in L-PBF, is in high demand across high-tech industries due to its exceptional strength-to-weight ratio and favorable casting properties. Despite these advantages, the mechanical characteristics and surface roughness of AlSi10Mg present challenges, which are further complicated due to the high thermal conductivity

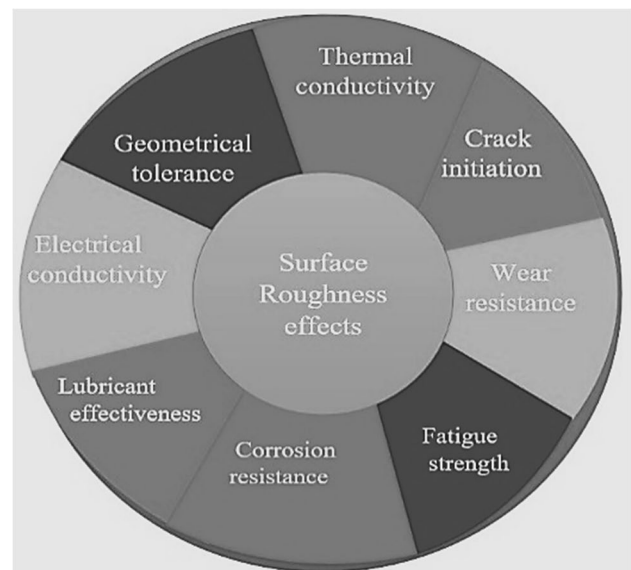


Fig. 1 Overview of product performance affected by surface roughness

and reflectivity of aluminum [29, 30]. Currently, several research and development efforts aim to overcome these challenges and achieve stringent quality standards for AlSi10Mg printed parts. The complex interaction of factors including material properties, thermal conductivity, and diverse process variables creates barriers to attaining consistent quality and reliability in the final components.

To address these challenges, machine learning introduces novel opportunities to model interactions and predict responses more accurately. For example, a study analyzed the effects of laser power, scanning speed, and hatch distance on surface roughness (Ra) using multilayer perceptron (MLP) and adaptive neuro-fuzzy inference system (ANFIS), with MLP demonstrating superior predictive capability [36]. Another investigation examined similar parameters, including layer thickness, utilizing an artificial neural network (ANN) with backpropagation, achieving an impressive R^2 of 99%, indicating high accuracy [37]. Linear regression, support vector machines, random forests, gradient boosting algorithms, and artificial neural networks are among the traditional machine learning algorithms commonly employed for predicting surface roughness in additively manufactured components [38]. According to Xu et al. [39], the effectiveness of machine learning models relies not only on the volume and accuracy of the data but also significantly on the choice of the modeling algorithm. Certain algorithms are particularly suited for modeling with limited data. The algorithms well suited for handling small datasets encompass support vector machines, Gaussian process regression, random forest, gradient boosting decision trees, eXtreme Gradient Boosting (XGBoost), and symbolic regression [39].

The L-PBF printing process involves numerous process parameters, each affecting the quality of the component in different ways. Considering all these parameters together to achieve specific quality characteristics is not efficient. It is crucial, therefore, to focus on key process parameters that most significantly influence the desired quality outcomes. Previous studies have explored various L-PBF effects of process parameters, such as laser power, scan speed, hatch spacing, and layer thickness, which significantly influence surface roughness [40–42]. However, the specific parameters considered in different studies vary, indicating that the literature on the effects of L-PBF process parameters on surface roughness is not yet mature.

In this study, we explored the interactions between L-PBF process parameters and surface roughness utilizing machine learning models to uncover underlying patterns and dependencies within the process. Particularly, random forest regression (RFR) and support vector regression (SVR) were considered based on the nature of the available dataset, quantity (small dataset), and capacity of the models to predict the surface roughness.

The study primarily focuses on AlSi10Mg, which was selected due to its significance in high-tech industries. The motivation for the study is the recognition of the critical importance of enhancing surface quality in L-PBF, as poor surface quality often leads to various premature failures. Moreover, it has been noted that the process parameters considered across different studies to analyze their effects on surface quality vary, highlighting the need for further investigation to systematically identify crucial parameters. Therefore, this study aims to contribute to the existing body of knowledge by formulating cause-and-effect mappings to pinpoint key process parameters responsible for surface roughness. It addresses the effects of these parameters on surface roughness and predicts their impacts using machine learning models, particularly RFR and SVR, followed by a comparison of their results to draw conclusions.

2 Materials and methods

2.1 AlSi10Mg powder

AlSi10Mg, an aluminum alloy containing aluminum, silicon, and magnesium, is a popular material for 3D printing due to its favorable mechanical properties, lightweight nature, and corrosion resistance. In this study, gas-atomized AlSi10Mg powder from m4p material solutions GmbH was used. The particle size distribution (PSD) of the powder was identified as D10 (23.41 μm), D50 (45.52 μm), and D90 (61.45 μm). Figure 2 shows (a) the particle size distribution and (b) a scanning electron microscope (SEM) image of the powder.

2.2 Parameter selection

To focus on the most significant factors and reduce the number of involved variables in the experiment, the relationship among the identified process parameters was thoroughly analyzed. As illustrated in Fig. 3, the cause-and-effect approach, guided by the “5Why?” principle, was implemented to identify the responsible process parameters for surface roughness formation. Accordingly, the detailed cause-and-effect analysis indicated that the laser power, scanning speed, layer thickness, and part orientation are the crucial parameters influencing surface roughness. Other parameters were held constant to streamline the experimentation process. Acknowledging the necessity of considering a full range of factors that may impact surface quality for a thorough understanding, the cause-and-effect analysis was derived mainly from literature and experience to limit the problem’s complexity.

The comparison presented in Table 1 put together the process parameters selected for sample fabrication in this study with those considered by other researchers in the

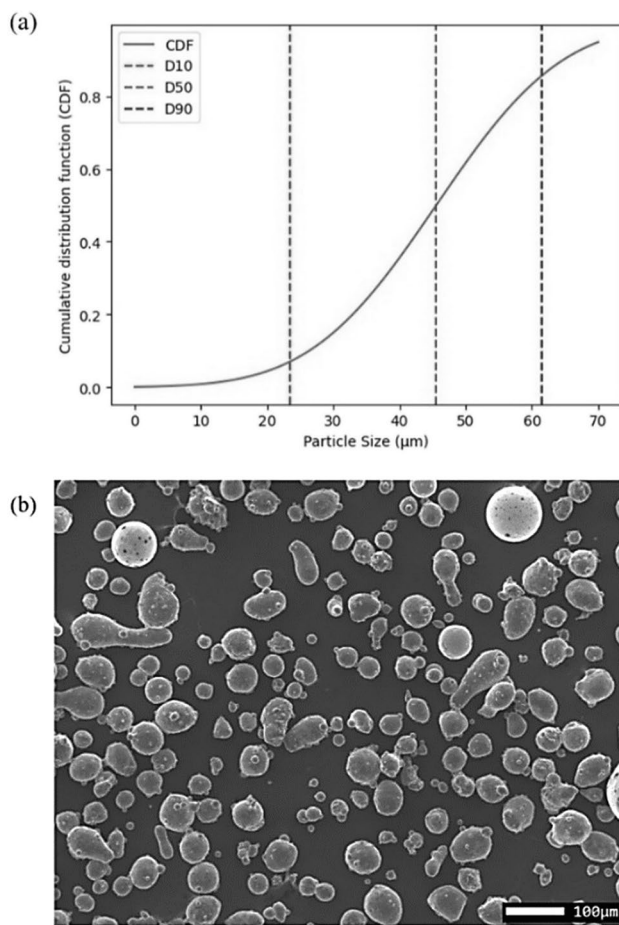


Fig. 2 a AlSi10Mg particle size distributions; b SEM image of the powder

manufacturing of AlSi10Mg using the L-PBF process, aiming to understand their impacts on various final product quality characteristics. This comparative analysis shows the distinctive perceptions offered by the current research. While previous studies have primarily focused on a constrained set of parameters such as laser power, scanning speed, and layer thickness, this study encompassed parameters such as build orientation and hatching distance. Such additional parameters are expected to enrich the analytical depth, giving valuable perceptions for optimizing surface quality in L-PBF.

2.3 Sample preparation and measuring

To investigate the effects of L-PBF process parameters on surface roughness, iFusion150 (Intech Additive Solution, Karnataka, 560058, India) was employed. The machine is equipped with two ytterbium fiber lasers, each possessing a maximum power of 500 W and scanning speeds of up to 7000 mm/s. The process chamber underwent a careful purge with 99.99% argon to ensure an oxygen-free environment. Powder storage was maintained in a vacuum desiccator to eliminate moisture infiltration, while the base plate was pre-heated to 200 °C to establish a controlled environment for the stored powder.

Cubic specimens, designed with precision at 15 mm×15 mm×15 mm using SolidWorks® CAD software (2023 SP0.1 professional version), were modeled adhering to ISO 4288:1997 [49] standards as depicted in Fig. 4.

Surface roughness measurements were conducted using a Mitutoyo Surftest SJ-301, with specified reference and evaluation lengths set at 2.5 mm and 12.5 mm, respectively, aligning precisely with guidelines [50]. Ahead of measuring the surface roughness, the surface was prepared to ensure it was free from any debris. The SJ-301 was

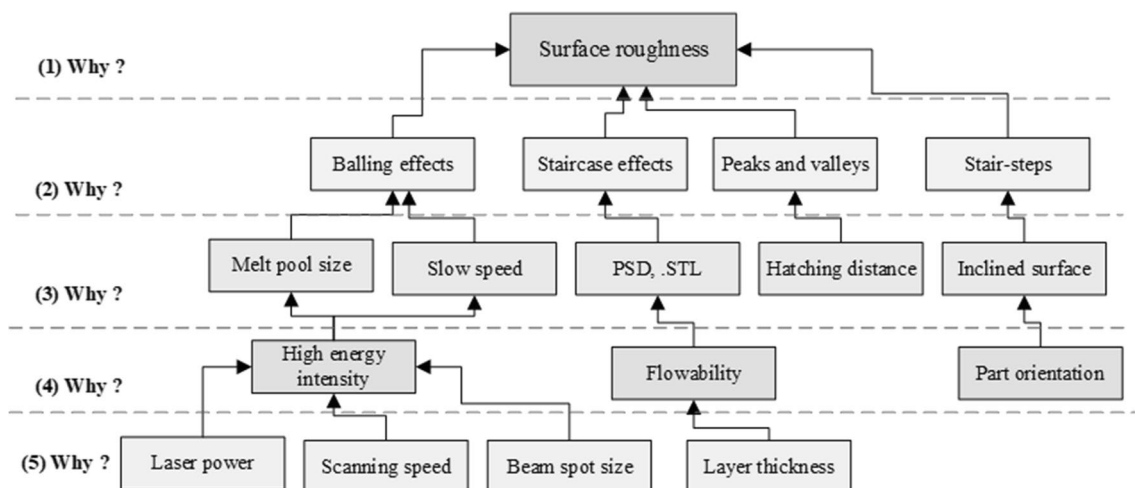
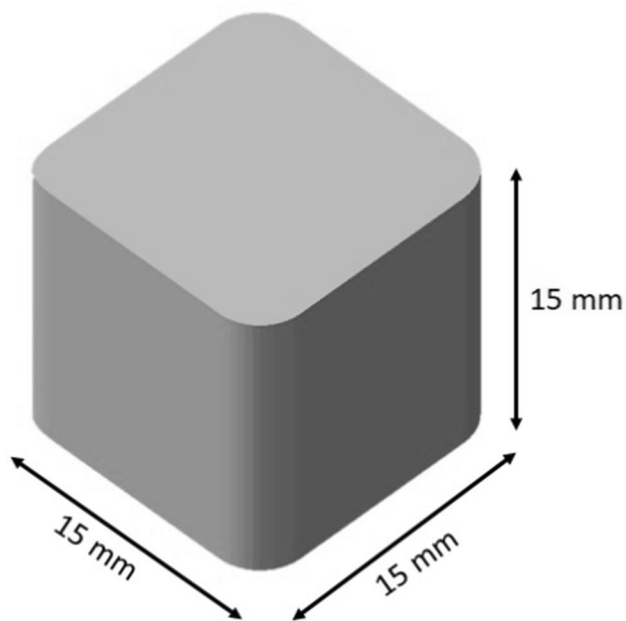


Fig. 3 Cause-and-effect analysis for surface roughness

Table 1 Comparison of selected process parameters considered for surface roughness study

Laser power (W)	Scanning speed (mm/s)	Layer thickness (μm)	Part orientation ($^\circ$)	Hatching distance (μm)	Additional parameter	Objective	[Ref.]
325, 350, 375, 400	600, 800, 1000, 1200	40, 60	0, 45	100	Build angle, speed & layer thickness	Surface roughness	In this study
320	900	30	–	80	–	Relative density	[43]
320, 360, 400	600, 750, 900	30	–	–	Overlap rate	Effect of heat treatment on relative density	[44]
320, 360, 400	600, 750, 900	30	–	–	Overlap rate	Surface roughness	[45]
200, 250, 300	800, 1000, 1200	20, 30, 50	–	100, 150, 200	Overlap rate	Hardness and roughness	[21]
350	1150	50	–	170	Bi-directional	Re-melting on surface roughness and porosity	[46]
350	1650	50	–	130	Contours	Surface quality and fatigue	[47]
350	1650	30, 50	0, 45, 90	100	Laser beam focus	Tribological properties	[48]

**Fig. 4** The 3D model of the specimens

then calibrated and set up according to the guidelines of the manufacturer, with parameters suitable for the surface roughness measuring standards. The probe was positioned perpendicularly to the surface to ensure proper contact, allowing it to traverse the surface and collect data on roughness parameters. The results of the measurement

are displayed digitally on the screen, with the option of printing results as depicted in Fig. 5.

2.4 Design of experiment

In this study, the design of experiment (DoE) according to the Taguchi method was employed to systematically examine the impacts of L-PBF process parameters on surface roughness. The Taguchi mixed design L16, outlined in Table 2, was chosen for its efficiency in analyzing multiple factors and their interactions while minimizing the number of experimental runs required [51–53]. To ensure reproducibility, triplicate samples were produced for each test case.

The samples were then fabricated according to the DoE with four levels of laser power (P) and scanning speeds and two levels of layer thickness (Lt) and part orientations (θ). As detailed in Sect. 2.1, all other parameters were maintained constant during sample production. Specifically, a bidirectional scanning strategy was utilized for its fast production capability, with a tilt angle of 67° and a hatching distance of $100\ \mu\text{m}$.

Statistical methods commonly used for surface roughness assessment include the metrics arithmetical mean (R_a) and root-mean-square (R_q) roughness parameters, which depict the vertical variances in surface height relative to a designated reference plane [50]. In this study, the roughness parameters according to ISO 4287, namely, the amplitude parameters (R_v and R_p) and the amplitude average parameters (R_a and R_q), were considered to characterize the surface roughness of AlSi10Mg prints.

Fig. 5 Surface measuring setup (Mitutoyo Surf Test SJ-301)

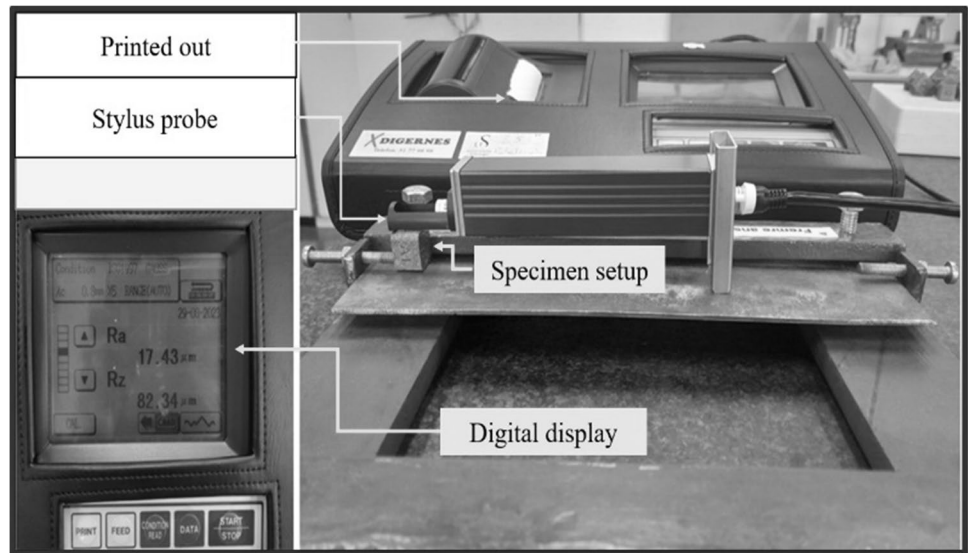


Table 2 The surface roughness results

Code	P (W)	V (mm/s)	L_t (μm)	θ ($^\circ$)	LED (J/mm)	VED (J/mm^3)	R_q (μm)	R_a (μm)	R_v (μm)	R_p (μm)
A1	325	600	40	0	0.54	135.4	10.29	7.92	-25.17	49.44
A2	325	800	40	0	0.41	101.60	10.15	8.05	-18.37	28.43
B1	325	1000	60	45	0.325	40.63	13.29	10.46	-36.72	33.86
B2	325	1200	60	45	0.27	33.85	13.97	10.88	-36.63	41.24
A3	350	600	40	45	0.58	145.8	10.23	8.06	-21.42	33.08
A4	350	800	40	45	0.44	109.4	12.33	9.16	-27.33	37.70
B3	350	1000	60	0	0.35	43.75	23.17	16.34	-59.78	52.48
B4	350	1200	60	0	0.29	36.46	22.51	18.17	-54.04	56.58
B5	375	600	60	0	0.63	78.13	14.93	11.53	-33.88	47.08
B6	375	800	60	0	0.47	58.59	13.82	9.79	-4.23	70.59
A5	375	1000	40	45	0.375	93.75	12.18	9.84	-24.18	36.13
A6	375	1200	40	45	0.31	78.13	12.86	9.99	-25.46	44.15
B7	400	600	60	45	0.67	83.33	22.41	18.24	-16.35	94.11
B8	400	800	60	45	0.50	62.50	26.15	19.68	-15.18	91.65
A7	400	1000	40	0	0.40	100.00	12.87	10.07	-24.36	40.34
A8	400	1200	40	0	0.33	83.33	13.46	10.43	-27.52	44.92

Amplitude average parameters, R_a and R_q , play pivotal roles in characterizing surface topography, especially in assessing the vertical aspects of surface deviations. R_a quantifies the arithmetic average of the absolute values of the profile height deviations from the mean line, measured within the evaluation length. According to ISO 4287, R_q is defined as the root-mean-square (RMS) deviation of the profile, capturing surface roughness by considering the square root of the mean of the squares of profile height deviations from the mean line. The determination and computation of R_a and R_q are depicted in Fig. 6, derived respectively from Eqs. 1 and 2. In these equations, $Z(x)$ signifies the profile height deviation from

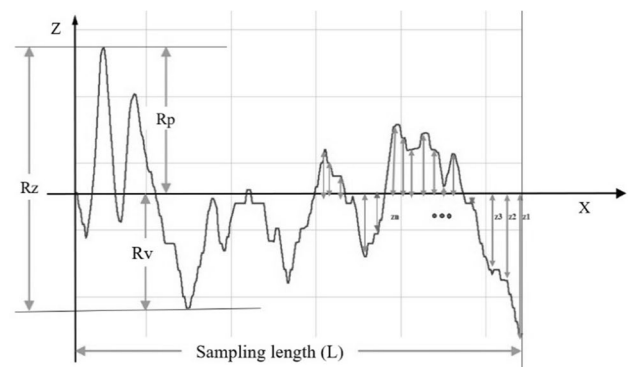


Fig. 6 Surface roughness parameter designation

the mean line at position X within the evaluation length, while X denotes the position along the surface profile within this length.

$$Ra = \frac{1}{L} \int_0^L |Z(x)| dx \quad (1)$$

$$Rq = \sqrt{\frac{1}{L} \int_0^L [Z(x)]^2 dx} \quad (2)$$

2.5 Developing machine learning model

Machine learning, a subclass of artificial intelligence (AI), encompasses various techniques that enable computers to learn from data and make predictions or decisions. These methods include the following:

- (i) Supervised learning, where models predict outcomes from labeled data
- (ii) Unsupervised learning, which identifies patterns in unlabelled data
- (iii) Reinforcement learning, where agents learn decision-making through rewards or penalties

The challenge of ML needs large data to make accurate predictions of complex relationships such as natural language processing, image recognition, and autonomous systems [54]. However, certain algorithms are well suited for modeling with limited data, including support vector machines, Gaussian process regression, random forest, gradient boosting decision trees, eXtreme Gradient Boosting (XGBoost), and symbolic regression [39]. In this study, the relationship between L-PBF process parameters and surface roughness was explored using machine learning models to uncover patterns and dependencies. Given the small dataset and the objectives, RFR and SVR were utilized, and their results were compared.

2.5.1 Data preparation

The data preparation process in these models included the data saved in a CSV (comma-separated values) file extracted. The variables representing features (laser power, scanning speed, layer thickness, and part orientation) and targets (Ra and Rq) were identified. To ensure consistency, feature scaling was performed by using *StandardScaler* from the *Sklearn.preprocessing* module. For predicting surface roughness, two machine learning algorithms (RFR and SVR) were employed. The dataset was divided into training (70%) and test (30%) sets for both models.

2.5.2 Model training

The FRF model is trained by the scaled features with hyperparameters including the number of trees, maximum depth, and minimum sample split optimized through cross-validation. The hyperparameters tuning was performed by using *GridSearchCV*.

Similarly, for the SVR model, the dataset was loaded and scaled by using standard *StandardScaler*, and the features were transformed to polynomial degree two. The hyperparameter optimization was carried out for the SVR model utilizing *RandomizedSearchCV*, where parameters including kernel type, regularization parameter, and kernel coefficient (gamma) were varied.

2.5.3 Performance metrics

In assessing model performance, key metrics such as the R^2 score (coefficient of determination), mean absolute error (MAE), and root-mean-square error (RMSE) were used (Eqs. 3 to 5). Model validation employed k -fold cross-validation ($k=5$) to ensure robustness and generalization to unseen data.

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (3)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (4)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (5)$$

where $\hat{y}$ denotes the predicted value for the i th observation, while y_i represents the actual value for the i th observation, $\bar{y}$ indicates the mean of the observed data, and n signifies the number of observations.

This approach facilitated effective training and hyperparameter tuning for optimal performance. MAE quantifies the average magnitude of errors irrespective of their direction, while RMSE offers a more comprehensive view by giving heavier penalties to larger errors due to squaring. R^2 measures the proportion of variance in the dependent variable predictable from the independent variables. By using k -fold cross-validation, the models underwent rigorous testing on various data subsets, boosting confidence in their performance across different scenarios and datasets.

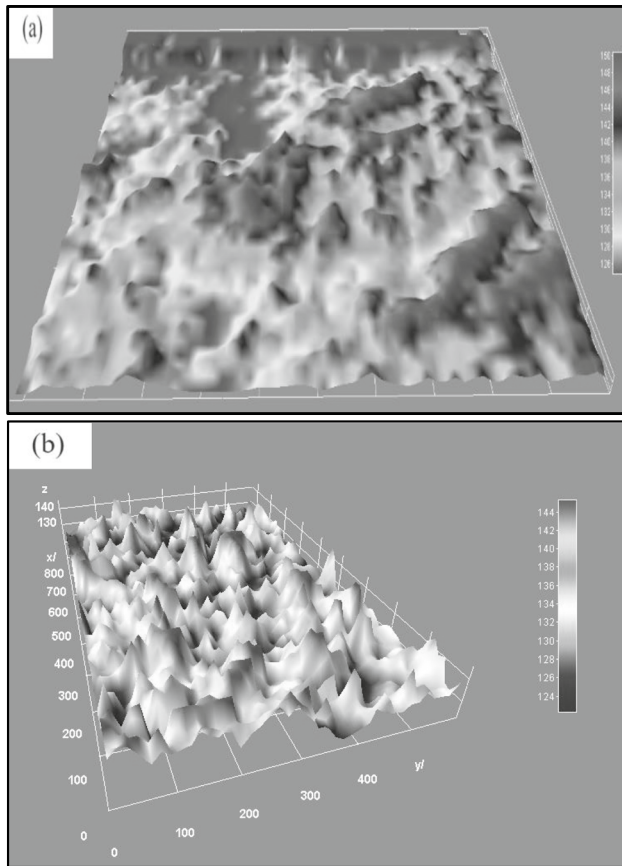


Fig. 7 Surface roughness at **a** lower laser power and speed and **b** higher laser power and speed

3 Results and discussions

3.1 Experimental results

The experimental results are summarized in Table 2. The results depicted in Fig. 7 illustrate the influence of key process parameters, as identified through cause-and-effect analysis, on surface roughness. The variations in laser power and layer thickness, particularly when combined with higher scanning speeds, consistently correlate with alterations in surface roughness across the examined power range.

For instance, the outcomes observed in two extreme cases, A1 and B8, are illustrated in Fig. 7a and b, respectively, showing the effects of parameters on surface patterns. In Fig. 7a, lower laser power and scanning speed contribute to a relatively smoother surface finish, particularly noticeable at lower layer thicknesses. However, the observation of unfused powder particles on the surface indicates incomplete fusion, potentially attributed to variations in PSD and their differential response to thermal effects. This suggests that while lower power and speed may improve surface finish, they can also lead to issues with incomplete material fusion.

On the other hand, Fig. 7b demonstrates that higher laser power levels lead to increased powder melting, which results in greater surface roughness due to the rapid formation of the melt pool. The significant influence of layer thickness is evident in terms of achieving proper fusion with the previous layer. This indicates that while higher power ensures better melting, it may poorly affect surface smoothness due to the aggressive thermal input.

These findings emphasize the interaction between process parameters and the resulting surface quality in L-PBF processes. The observed trends highlight the importance of parameter optimization, as balancing these parameters is crucial to achieving the desired surface characteristics. This discussion emphasizes the necessity of process control in L-PBF to mitigate surface roughness while ensuring complete material fusion.

Main effect plots are utilized to show the contribution of individual process parameters, assessing their significance in causing variations in surface roughness. These analyses aid in determining which parameters employ the greatest influence on surface roughness and exposing how alterations in these parameters impact the quality of printed parts. The impacts of each level of process parameters on surface roughness are depicted in Fig. 8, where a *smaller is better* signal-to-noise ratio analysis was conducted. The main effect plot shows low values for Ra and Rq results (Fig. 8a and b) implying improved surface roughness. However, better results for Rv (Fig. 8c) were observed at high laser power, medium scanning speed, low layer thickness, and inclined build angle. Similarly, better Rp results (Fig. 8d) were obtained at low laser power, high scanning speed, low layer thickness, and inclined part orientation. The improved results for Rq were achieved at low laser power, scanning speed, thickness, and flat build angle. The results in Table 3 offer additional insights into the degree of influence contributed by each selected process parameter, with consideration given to both Ra and Rq. Layer thickness proves most influential, closely followed by laser power, with scanning speed and build angle ranking third and fourth, respectively.

3.2 The effects of energy density on surface roughness

Linear energy density (LED) is a parameter that represents the amount of energy delivered per unit length of a laser scan path, while volumetric energy density (VED) in L-PBF refers to the amount of energy delivered to a unit volume of the powder bed during the melting process. It is typically measured in joules per millimeter. Examining the impact of LED and VED reveals their significant influence on surface roughness parameters (Ra, Rq, Rv, Rp). The LED (J/mm) and VED (J/mm^3) are calculated by using Eqs. 6 and 7, respectively.

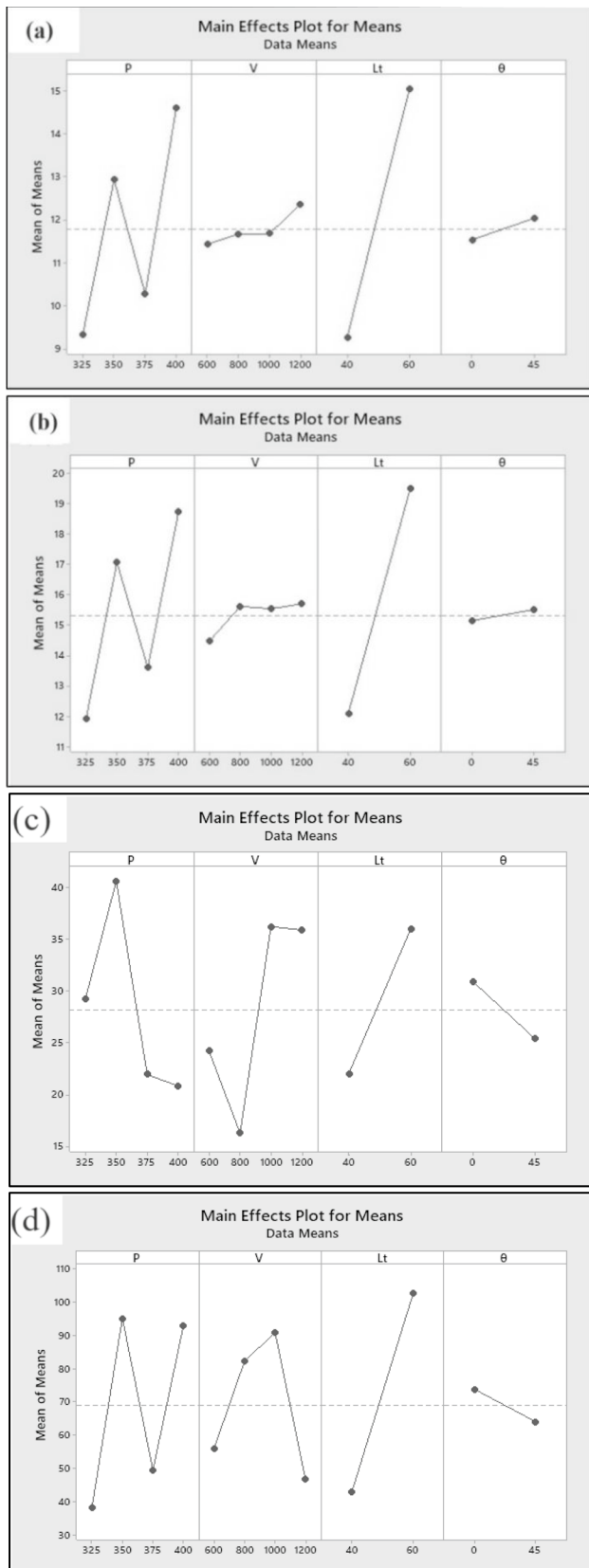


Fig. 8 Main effect plot for **a** Ra, **b** Rq, **c** Rv, and **d** Rp

Table 3 Signal to ration smaller is better: level of significance on surface roughness

Level	P	V	Lt	θ
1	−21.44	−22.74	−21.59	−23.22
2	−24.09	−23.28	−25.51	−23.39
3	−22.66	−23.53	0	0
4	−25.03	−23.68	0	0
Delta	3.59	0.94	3.92	0.16
Rank	2	3	1	4

$$\text{LED} = \frac{P}{V} \quad (6)$$

$$\text{VED} = \frac{P}{V \times h \times \text{Lt}} \quad (7)$$

where P = laser power (watt), V = scanning speed (mm/s), h = hatching distance (mm), and Lt = layer thickness (mm).

The outcome provides valuable insights into the relationship between energy inputs and surface roughness in L-PBF manufacturing processes. As depicted in Fig. 9, increasing LED corresponds to higher Ra and Rq, indicating a rougher surface profile. For instance, adhering to Table 2, at 0.5 J/mm^2 , the Ra and Rq are 19.68 and $26.15 \text{ } \mu\text{m}$, while the values at 0.27 J/mm^2 are 13.97 and $10.88 \text{ } \mu\text{m}$, respectively. The positive correlation suggests that higher LED values result in increased overall roughness. The impact of LED on Rp is less clear, displaying a varied pattern without a consistent trend. In the case of low LED, the laser may not provide enough energy to fully melt and fuse the powder particles. Insufficient melting can result in poor adhesion between powder particles and layers, i.e., leading to a rough surface finish, while higher energy leads to excessive melting, spreading, and coalescence of the molten material. Similarly, increasing VED is associated with varied effects on surface roughness parameters. The Rv values in both LED and VED are complex and exhibit both increasing and decreasing trends, suggesting additional influencing factors. The irregular patterns of Rv and Rp are associated with issues such as balling, spattering, or the formation of large pores, which the results manifested on the surface. At higher VED values, the Ra and Rq values become stable and tend to decrease overall roughness. Higher VED led to better material consolidation and melting, which may contribute to reduced surface roughness. Even though factors such as PSD can also affect energy absorption and distribution, thereby influencing both volumetric energy density and surface roughness, low VED results have incomplete melting and poor surface quality.

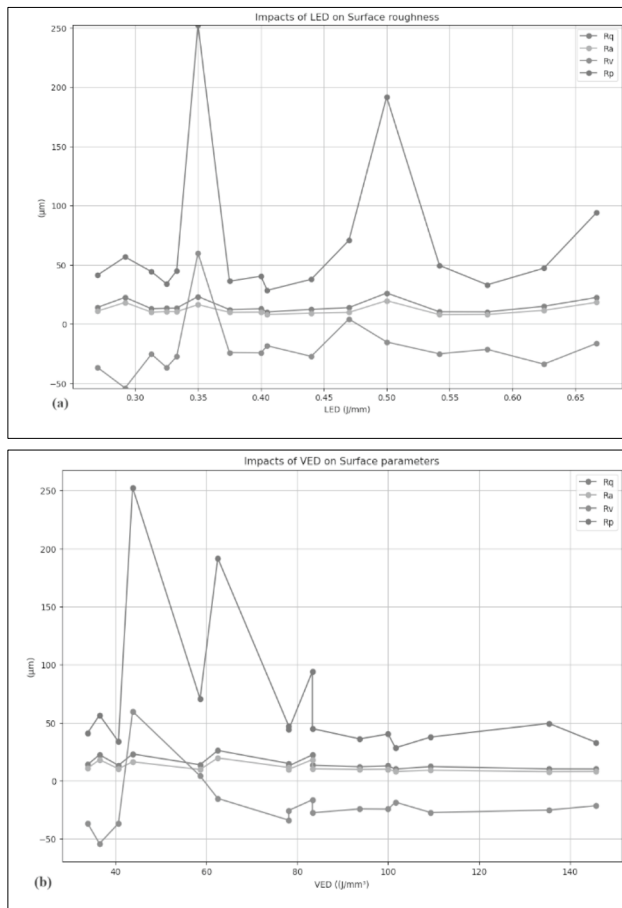


Fig. 9 Impacts of **a** LED and **b** VED on surface parameters

3.3 Surface characteristics

During the L-PBF process, various surface characteristics and defects significantly influenced the quality of printed parts, including spattering (fine particle ejection) and balling (undesirable clump formation), which impact the quality of printed layers. Spattering occurs due to high-energy laser interactions, leading to the expulsion of molten material [55]. Additionally, porosity, characterized by voids within the material, may result from inadequate powder fusion or gas entrapment [56]. Blistering, characterized by bubble-like structures, is often linked to gas entrapment during melting. Many of these common defects manifest on the surface, thereby influencing surface roughness parameters, as depicted in Fig. 10.

Balling and spattering are observed during the L-PBF process, particularly under conditions of elevated laser power. Low laser power levels contribute to incomplete melting and poor fusion of powder particles, resulting in the formation of unconsolidated clusters and irregularities on the surface, as evidenced by increased R_p (peak height) and R_v (valley depth) values. Moreover, specific conditions

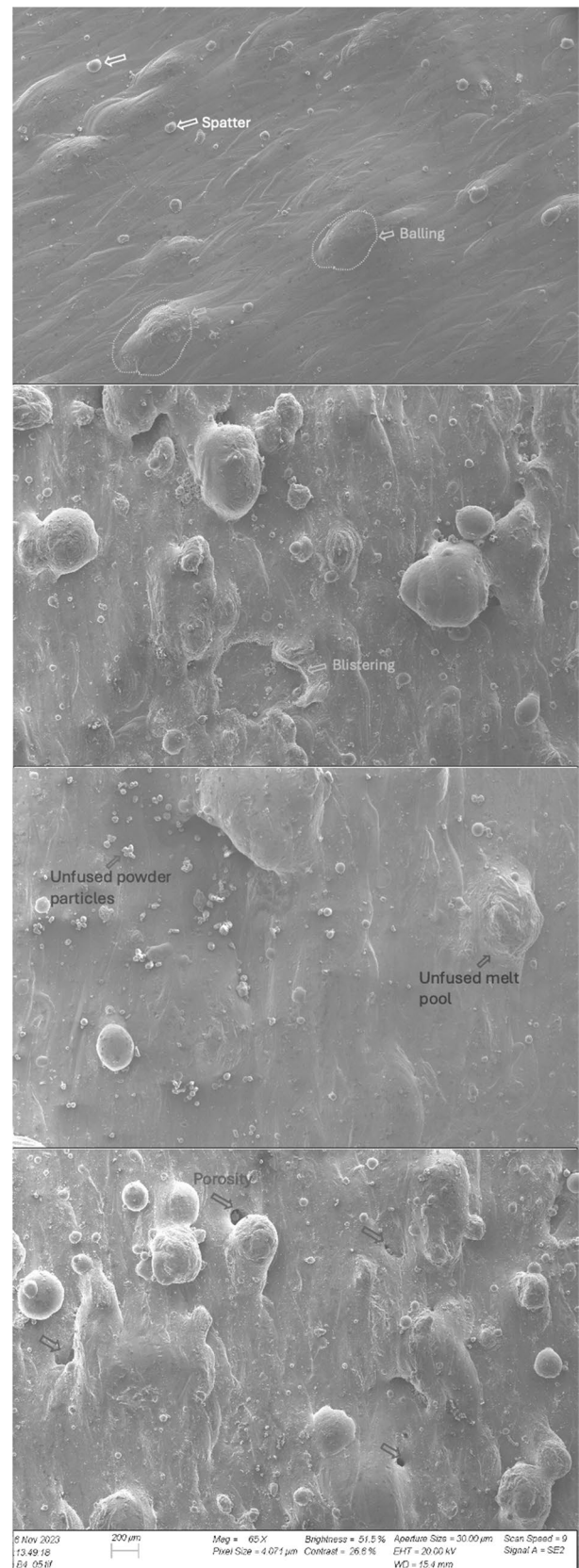


Fig. 10 Surface characteristics of AlSi10Mg samples printed by L-PBF

during the L-PBF process may induce spattering, leading to the splattering of molten material and further affecting surface quality. Porosity, blistering, and unfused melt pools are common consequences, presenting as irregularities, voids, and incomplete fusion. These defects are exacerbated by factors such as high scanning speed, increased layer thickness, and build angle, collectively heightening the risk of surface quality issues.

3.4 Exploring surface roughness prediction

3.4.1 Random forest regression

Random forest regression is a machine learning method that constructs numerous decision trees from random subsets of data and features. It improves accuracy and minimizes overfitting by averaging the outputs of these trees, resulting in a robust and dependable model.

The RFR models employed in this study predict the effects of key process parameters on surface roughness (Ra and Rq). The performance of these models was evaluated by using MAE, RMSE, and R^2 . The RFR model for Rq yielded $MAE=1.71$, $RMSE=2.11$, and $R^2=0.85$, while the performance for Ra was better with $MAE=0.51$, $RMSE=0.66$, and $R^2=0.97$. These metrics indicate that the model predicts Ra better than it does Rq, as lower error values and higher accuracy were achieved.

Moreover, the visual analysis through predicted vs. actual values was conducted to complement the numerical evaluation. As illustrated in Fig. 11, the alignment between the model predictions and actual values provides valuable indications of the model performance. The comparison between predicted and actual values allows for a direct examination of how well the model's predictions correspond to the ground truth. These visualizations assist in determining the suitability of the model for making predictions on new or unseen data.

3.4.2 Support vector regression

Support vector regression is a robust algorithm designed for regression tasks, demonstrating proficiency in handling nonlinear data by optimizing hyperplanes to minimize error margins rather than focusing solely on precise data fitting. Its efficacy stems from its utilization of kernel functions like linear, polynomial, or radial basis functions to transform data into higher dimensions, accommodating the complexities inherent in various datasets. This adaptability enhanced by SVR to adjust has the capability to address both linear and nonlinear relationships, facilitated by hyperparameters such as regularization and kernel parameters.

This study focused on the relationship between L-PBF process parameters and surface roughness, employing

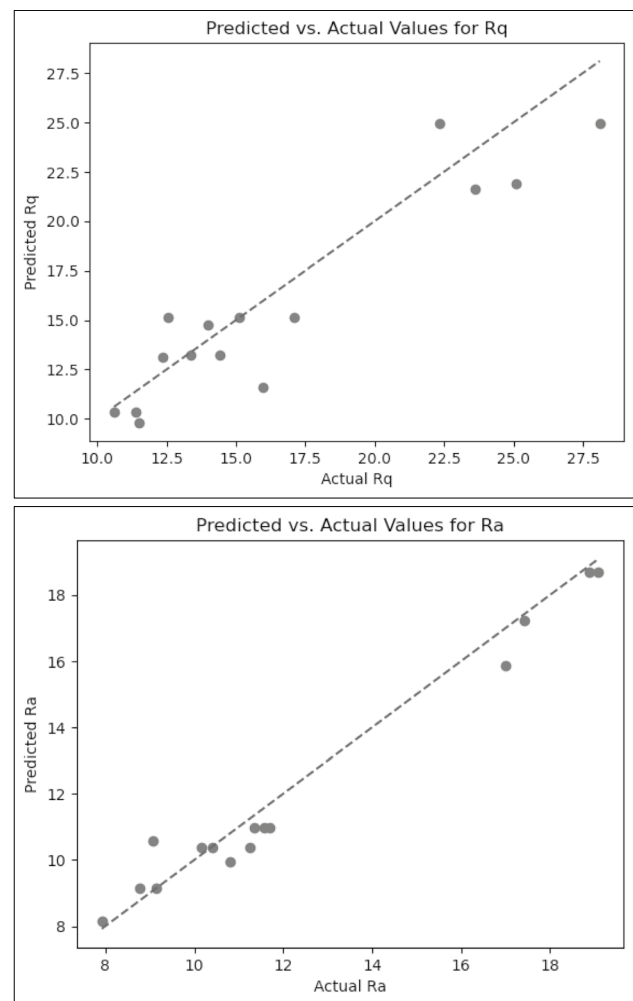


Fig. 11 Performance validation of RFR model

machine learning to show underlying patterns and dependencies. Due to the limitations posed by the small dataset size, the study opted for a dual approach, utilizing both SVR and RFR in parallel to predict surface roughness parameters. This strategic combination allowed us to pull the strengths of each algorithm, ensuring a robust and comprehensive analysis tailored to the intricacies of our dataset and research goals.

The aim was to enhance the predictive performance of SVR models for surface roughness prediction by employing several optimization strategies. Initially, feature engineering was implemented by adding polynomial features to capture potential nonlinear relationships between input features and target variables. A grid search for optimal hyperparameters was conducted using *GridSearch* exploring various combinations of the regulation parameters, kernel type, and gamma value. However, to address computational constraints and explore a wider hyperparameter space more efficiently, a transition was made to *RandomizedSearchCV*. This approach

allowed for a more extensive search within the defined computational range, yielding improved combinations of hyperparameters. Moreover, to power the strengths of multiple SVR models, an ensemble technique called averaging was applied, combining predictions from SVR models trained with different random seeds. These enhancements are aimed to improve the accuracy and robustness of the SVR model in predicting surface roughness.

The performance of the model was evaluated using MAE, RMSE, and R^2 . The SVR model for Rq resulted in an MAE of 2.4, RMSE of 3.1, and R^2 of 0.66, while for Ra, it performed better with an MAE of 0.99, RMSE of 1.41, and R^2 of 0.85. Similar to the case of RFR, the results of the metrics indicate that the model predicts Ra better than it does Rq, as lower error rates and relatively better accuracy are obtained.

The SVR model demonstrates divergent performance on the dataset, as evidenced by metrics values such as $R^2=85\%$ for Ra, indicating better performance, while $R^2=66\%$ for Rq, which is moderate. This difference in model performance is visually agreed by the plotted data points in Fig. 12, which align closely with the metrics findings. Particularly, the plot reveals a systematic trend in Rq predictions, with most data points falling below the regression line, suggestive of a tendency for the model to underestimate Rq values. This trend corresponds well with the lower R^2 value observed for Rq,

indicating a less comprehensive capture of variance compared to Ra. In the case of Ra, the plot depicts relatively a tighter alignment between predicted and actual values, reflecting a higher R^2 value and implying a more accurate interpretation of the underlying data relationships. This combined analysis indicates the significance of both numerical and visual assessments to evaluate the developed model.

3.4.3 Comparison the models

In this study, the effects of the L-PBF process parameters on surface roughness were predicted by the two models, namely, RFR and SVR. After careful hyperparameter optimization, the models' performance was assessed using the R^2 , MAE, and RMSE, and the results are given in Table 4. The RFR model outperformed the SVR model in predicting both Ra and Rq. This indicates that the RFR model was more effective in capturing data variance and delivering more accurate predictions. Similarly, for Ra, the RFR model exhibited superior performance, significantly higher than the SVR model, and showed lower error metrics, reflecting more reliable and precise predictions.

The analysis of the graphical plots (Figs. 11 and 12) confirmed that the RFR model outperformed the SVR model. The RFR model showed strong alignment between predicted and actual values for both Ra and Rq, indicating high accuracy and low error rates. The SVR model tended to predict lower Rq values, and while it performed better for Ra, it still did not match the accuracy of the RFR model.

Overall, the RFR model proved to be more robust and reliable for predicting surface roughness in this context.

4 Conclusions

This research investigates the effect of L-PBF process parameters on surface roughness through employing machine learning techniques. L-PBF, a prominent metal-based AM technique, offers manufacturing flexibility and complex design possibilities. However, some challenges related to surface roughness in L-PBF-printed parts, particularly for applications in high-tech industries, remain a critical concern. This study is aimed to contribute to the existing knowledge on the identification of the critical

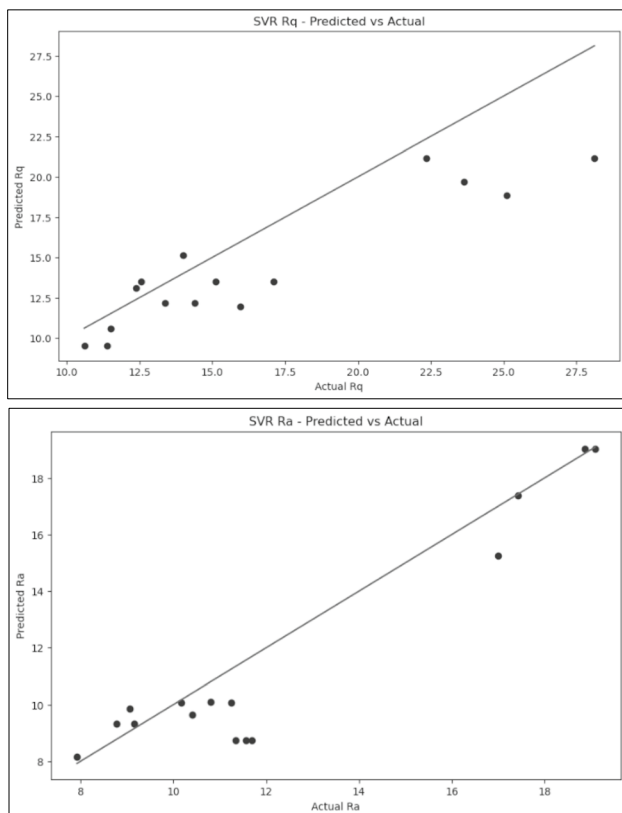


Fig. 12 Performance validation of SVR model

Table 4 Performance comparison of RFR and SVR models

Metrics	RFR		SVR	
	Ra	Rq	Ra	Rq
MAE	0.5	1.71	0.99	2.4
RMSE	0.66	2.11	1.41	3.1
R^2	0.97	0.85	0.85	0.66

parameters that influence the surface quality of parts fabricated by L-PBF. Based on this study, the following conclusions are drawn.

The cause-and-effect analysis results were proven, confirming that the identified factors (build angle, scanning speed, and layer thickness) indeed have a significant impact on the final product's surface finish.

Layer thickness proves most influential, closely followed by laser power, with scanning speed and build angle ranking third and fourth, respectively.

The findings demonstrate that higher LED values correlate with increased amplitude average parameters (Ra and Rq), indicating rougher surfaces. Insufficient LED can lead to poor adhesion and rough surfaces, while excessive energy causes excessive melting. Higher VED tends to stabilize and decrease Ra and Rq, indicating improved material consolidation and reduced overall roughness.

The correlation observed between amplitude parameters (Rv and Rp) and energy density input (LED and VED) suggests the presence of additional influencing factors, thereby necessitating further investigation.

The machine learning models, specifically RFR and SVR were employed to predict the effects of these process parameters on surface roughness. The RFR model achieved R^2 values of 97% and 85% for predicting Ra and Rq, respectively, while the SVR model achieved 85% and 66%, respectively. Evaluation metrics demonstrated that the RFR model outperformed the SVR model in predicting surface roughness, which implies that FRF model is more effective in capturing data variance and delivering better prediction with the dataset.

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Code availability Not applicable.

Declarations

Ethical approval Not applicable.

Consent to participate Not applicable.

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